



# Technologies for Digitalization



1. Cloud computing: Intro and Infrastructure as a Service (IaaS) – 3 weeks
2. Cloud Computing: Container as a Service (CaaS) – 4 weeks
3. Cloud Computing: Infrastructure as Code (IaC) – 1 weeks
4. Cloud Computing: DevOps – 3 weeks
5. Cloud Computing: Platform as a Service (PaaS) – 1 week
6. Cloud Computing: Function as a Service (FaaS) – 1 week
7. Cloud Computing: Review– 1 day
8. Cloud Computing: Project– 1 day



## Unit 1

- Introduction to Cloud Computing
- Technology Enablers
- Cloud Platforms
- Automation and Orchestration
- Cloud Security



# Introduction to Cloud Computing

Professor: Rogelio Martinez Perea  
rmperea@comillas.edu



- What is Cloud Computing
- The history of Cloud Computing
- Cloud Service models
- Cloud deployment models
- Why Cloud Computing



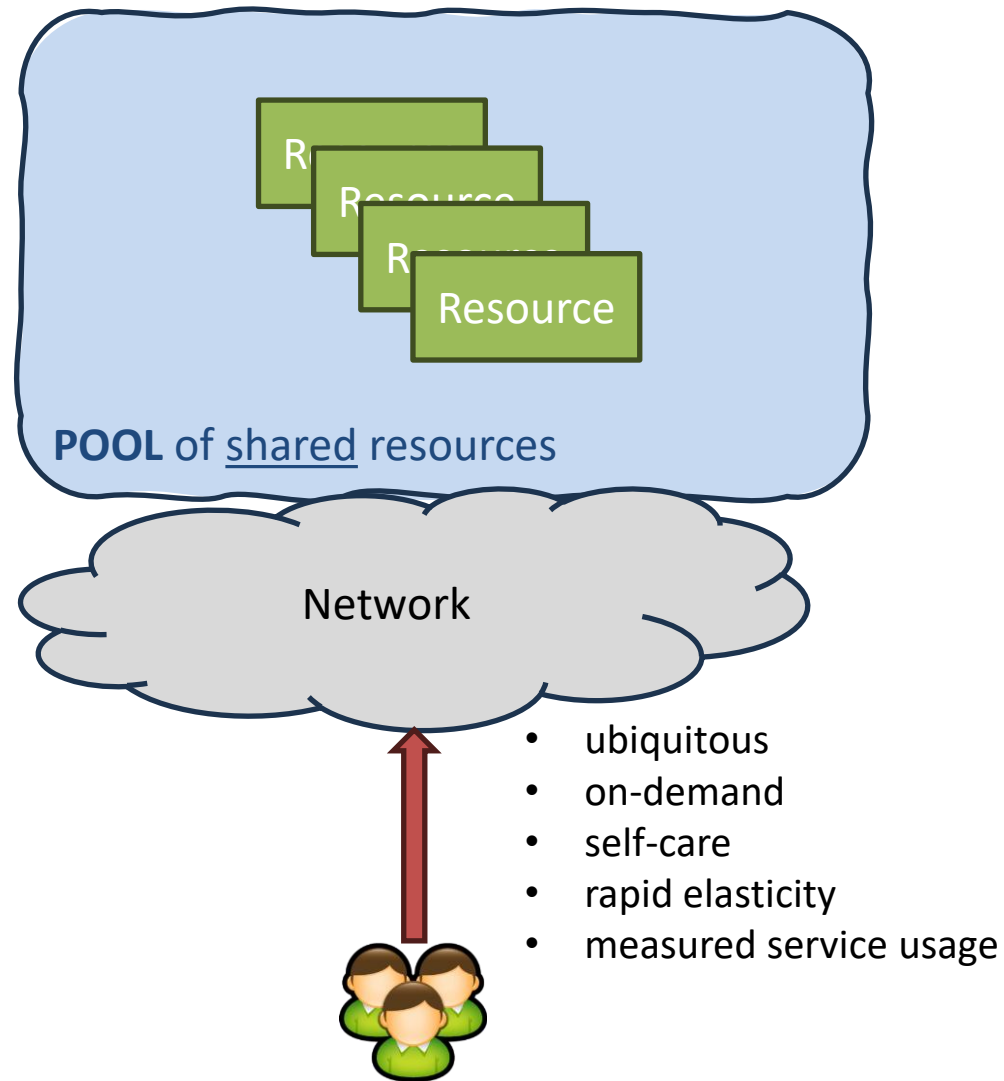
# What is Cloud Computing



“a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.” (\*)

(\*) Peter Mell, Timothy Grance: “The NIST Definition of cloud computing”, NIST Special Publication 800-145

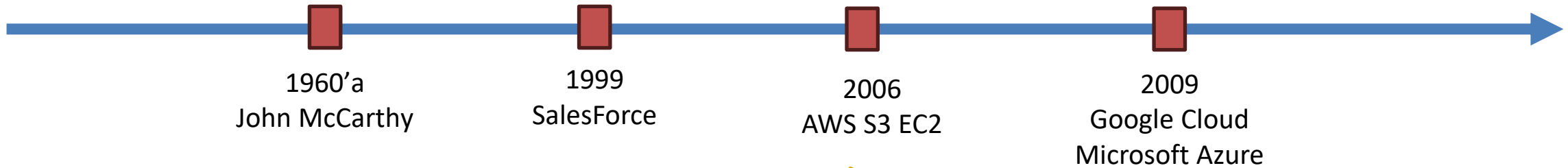
# Main aspects







# History of Cloud computing



- 2000 e-commerce company with scalability problem
- 2000 help 3<sup>rd</sup> party merchant to leverage e-commerce platform
  - build platform to manage hyper growth → made them very efficient . API'fication → built a common platform
- 2003 High skills at running reliable, scalable ad cost effective data centers → others might benefit
- 2006 S3 and EC2. Internet as Operating System
- Today: over 200 services, 34 geographical regions, 90 billion dollars revenue, leader in market



# Service models

# Main Cloud Service models



## IaaS

- the cloud service delivers infrastructure resources to organizations via the cloud, such as compute (virtual machines), storage, and networking.
- customers don't have to manage, maintain, or update their own data center infrastructure, but are responsible for configuring and maintaining operating systems, applications, etc.

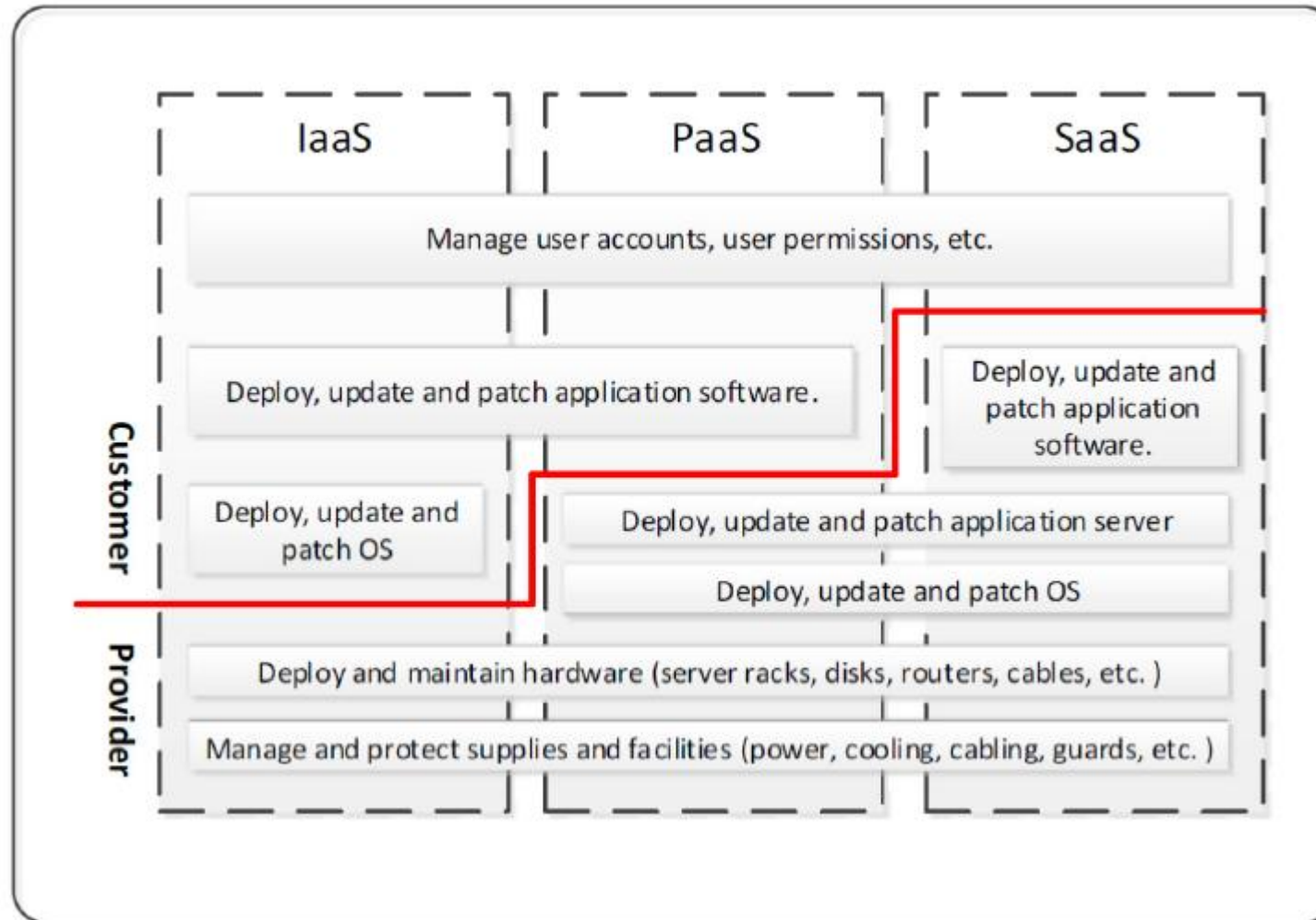
## PaaS

- the cloud service provider delivers and manages all the hardware and platform resources to develop applications through the cloud.
- customers can use PaaS to develop, run, and manage applications without having to build and maintain the infrastructure or platform on their own. Customers must write the code and manage their data and applications, but the environment to build and deploy apps is managed and maintained by the cloud service provider.

## SaaS

- the cloud service provider delivers the entire application stack, delivering an entire cloud-based application that customers can access and use. SaaS products are completely managed by the service provider and come ready to use, including all updates, bug fixes, and overall maintenance. Most SaaS applications are accessed directly through a web browser, which means customers don't have to download or install anything on their devices.

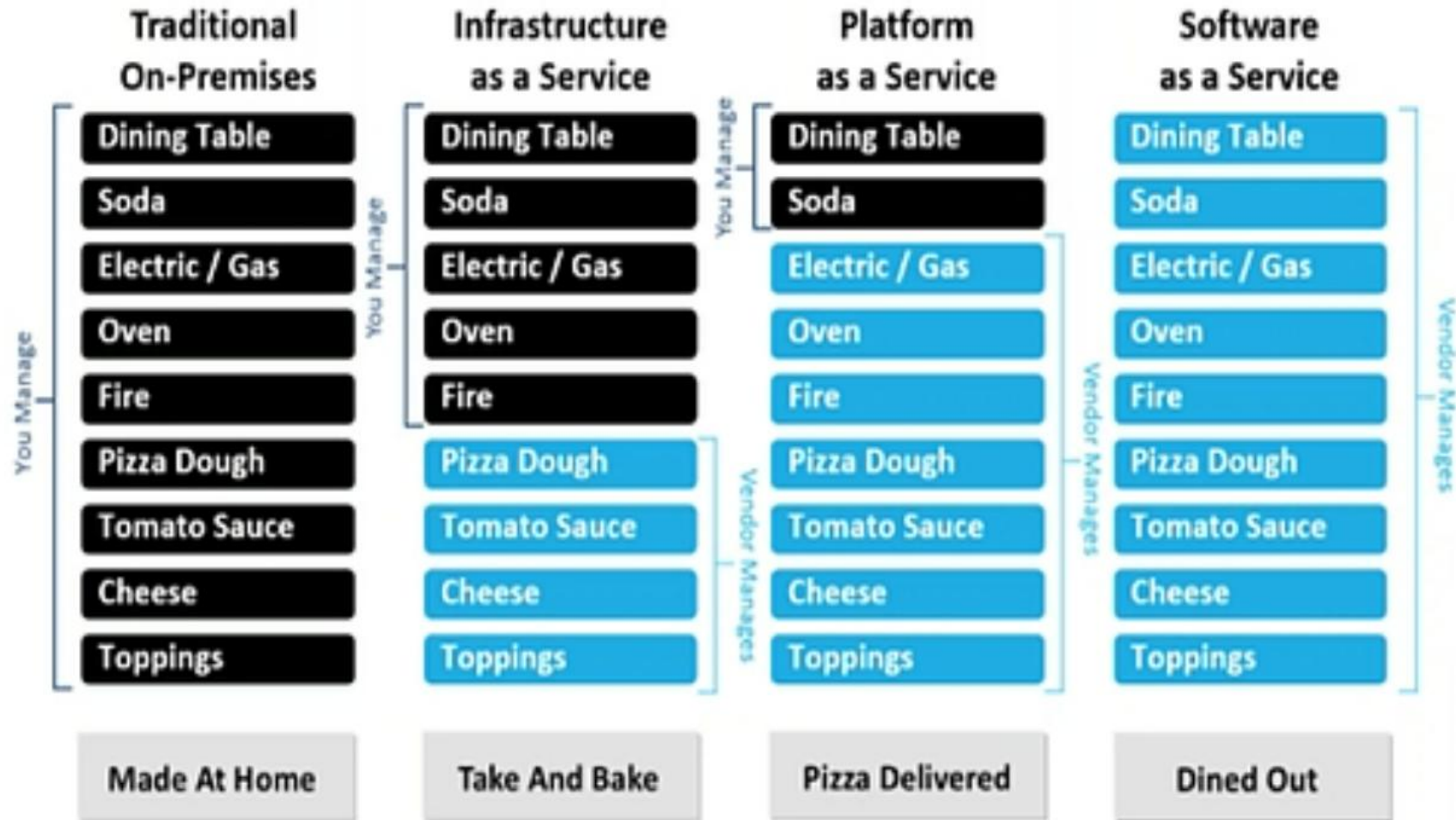
# Main service models comparison



(\*) 5Dr. M.A.C. Dekker, Dimitra Liveri: "Cloud Security Guide for SMEs", European Union Agency for Network and Information Security



## The Pizza & The Cloud – How Many Ways?



(\*) Cloud1: <https://www.securecloudaas.com/making-sense-of-all-the-cloud-as-a-service-models/>



## CaaS

- the cloud service provider delivers and manages all the hardware and software resources to develop and deploy applications using containers
- customers can use CaaS to develop, run, and manage applications without having to build and maintain the infrastructure or platform to run and manage containers. Customers must write the code and manage their data and applications

## FaaS

- (aka serverless) the cloud service provider delivers a platform to build applications as simple, event-triggered functions without managing or scaling any infrastructure



# Cloud deployment models



# Cloud deployment models



## Public

Cloud Service Provider



Customer1



Customer2

## Semi-private

Cloud Service Provider



Customer1



Customer2

## Private



Customer1 =  
Cloud Service Provider

## Hybrid

Cloud Service Provider



## Community



## Multi

Cloud  
Service  
Provider 1



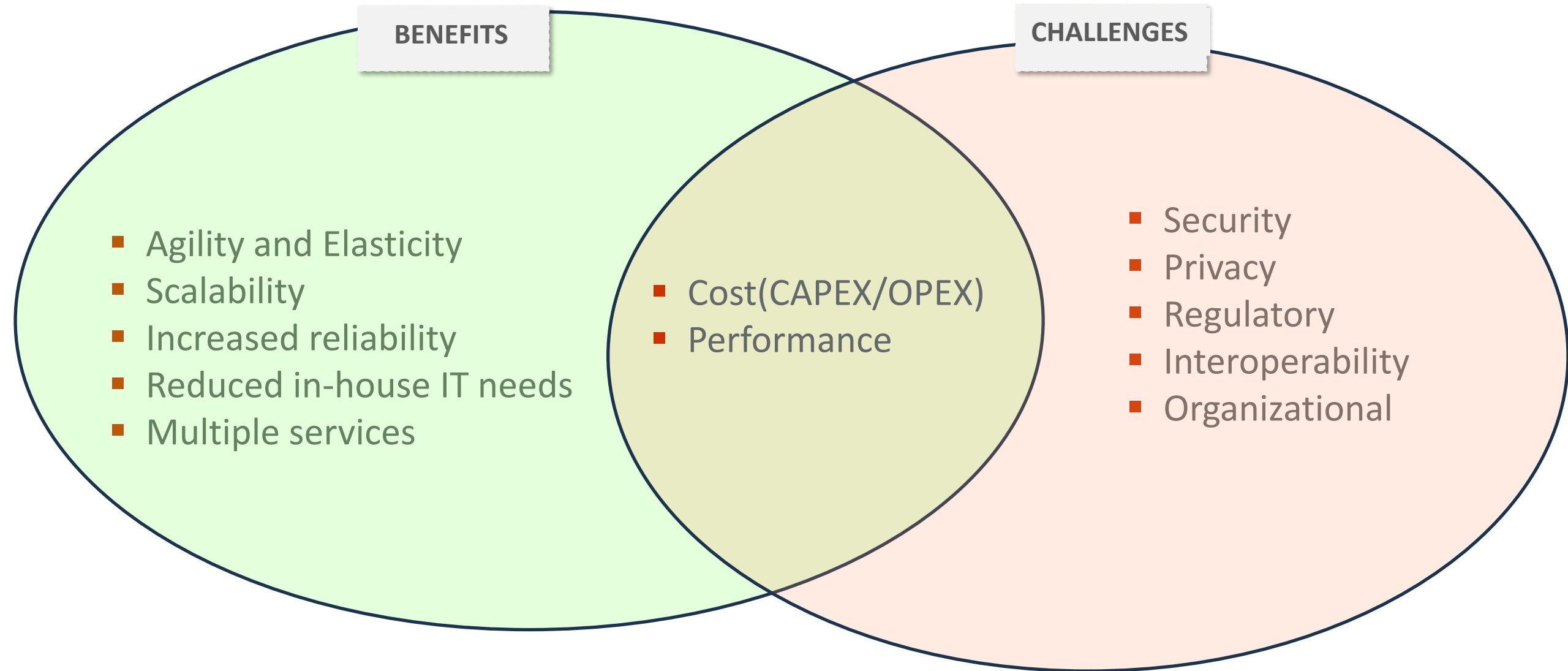
Cloud  
Service  
Provider 2



Customer1



# Why Cloud computing





# Technology enablers

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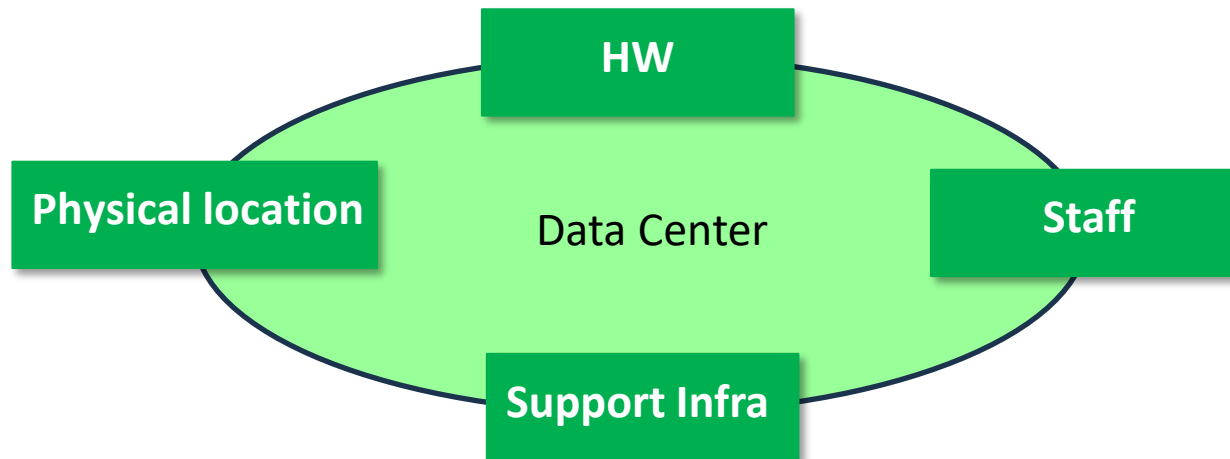


- Data center
- High speed networking
- Virtualization

# What is a data center



- Physical facilities that host computer systems and their associated components to store, process, manage, and distribute large amounts of data





- Enterprise Data Centers
- Colocation Data Centers
- Managed Data Centers
- Cloud Data Centers
- Edge Data Centers





- Physical Security
- Power
- Climate control
- Network connection
- Fire suppression systems
- Server organization







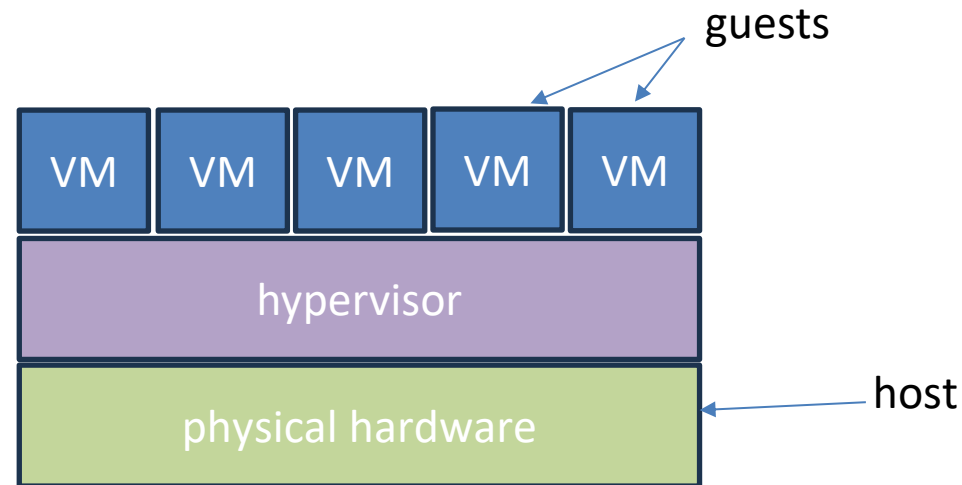
# Virtualization



- Virtualization is the creation of a virtual version of a technology resource, such as a server, a storage device or a network resource
- The main idea is that computing resources can be virtually partitioned into several virtual machines.
- Main benefits:
  - it can maximize the hardware utilization
  - agile and dynamic creation of resources
  - more flexible and agile management of infrastructure (e.g. hardware upgrades, hardware issues, relocation of VMs)
  - simplifies and harmonizes the hardware infrastructure in the data center



- It is achieved through a piece of SW called the hypervisor
- The hypervisor creates and runs virtual machines (VMs)
- The hypervisor treats CPU , memory and storage resources as a pool that can be easily reallocated between existing VMs or to new VMs
- The hypervisor gives each virtual machine the resources that have been allocated and manages the scheduling of VM resources against the physical resources.





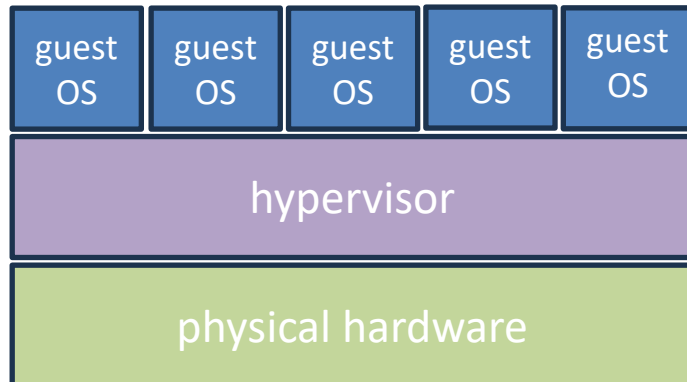
- The hypervisor has access to and manages all the hardware resources.
- A guest operating system and the application programs on top are then managed under control of the hypervisor.
- When a guest OS performs certain operations, such as a privileged instruction that directly involves the shared hardware resources, the operation is intercepted by the hypervisor and performed by the hypervisor on behalf of the guest.
- Server virtualization provides strong isolation between the VMs and encapsulation.
- Guest software is unaware of the work performed by the hypervisor.

# Hypervisor types



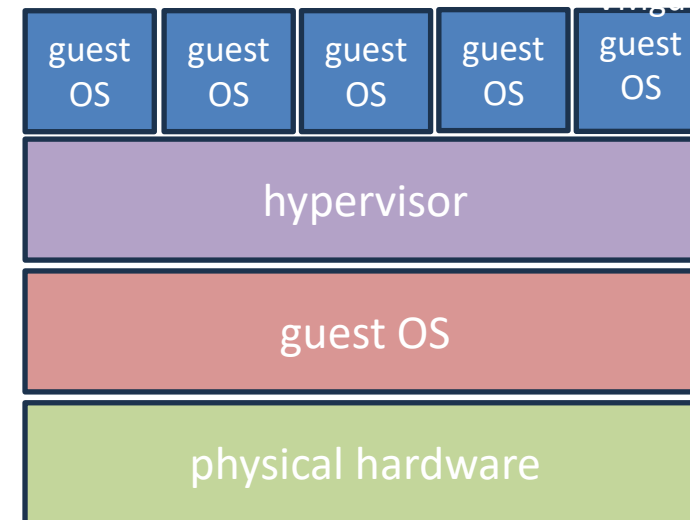
## Type 1

- It runs directly on the HW and the VMs sit on top
- It can achieve higher levels of availability, security and performance.



## Type 2

- It runs on top of an existing operating system
- It does not have direct control of the underlying HW
- Less performant than Type 1

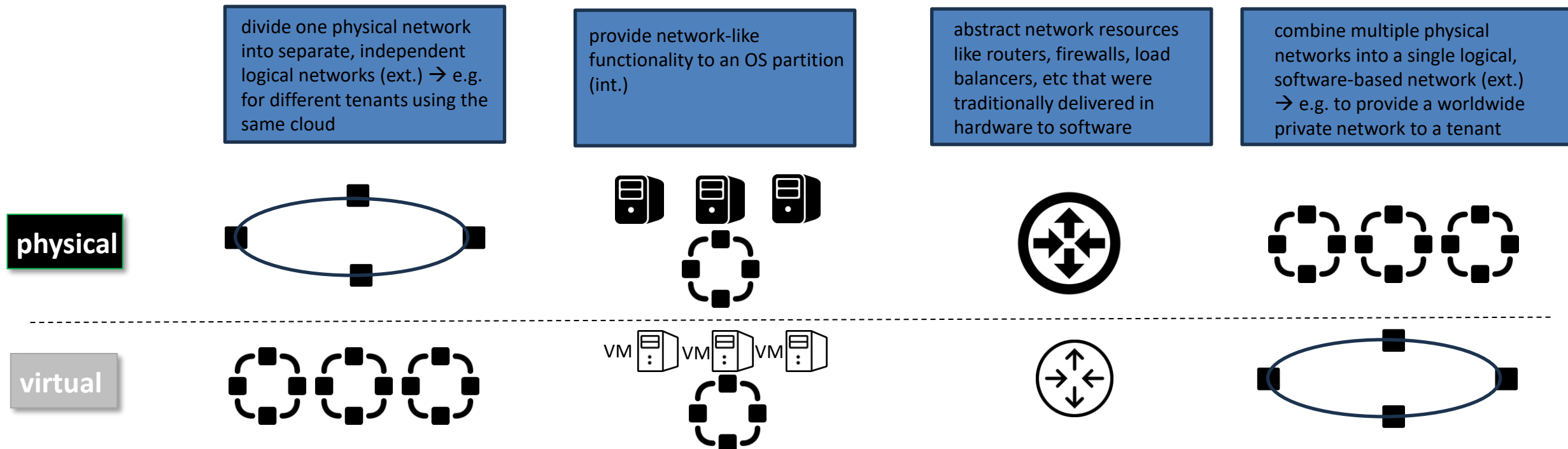


# Network virtualization



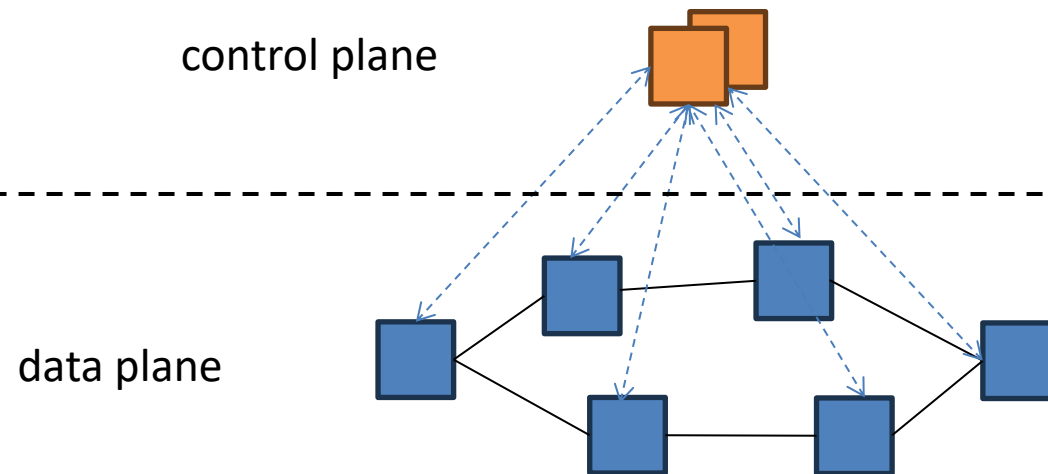
## PURPOSE

- **Support VM networking** | Provide vNICs, vSwitches, overlays, routing, isolation, and mobility.
- **Enable multi-tenant isolation** | Each customer gets a private virtual network.
- **Provide virtual network services (NFV)** | Firewalls, load balancers, VPNs, routers — all in software.
- **Allow rapid provisioning and automation** | Networks created in seconds, not days.
- **Improve scalability** | Millions of virtual networks on one physical fabric.
- **Strengthen security** | Micro-segmentation, per-VM firewalls, consistent policies.
- **Simplify operations** | Via logically centralized control





- It may refer to different but related concepts, but which share the key idea of decoupling the control plane and the data plane
- When combined with network virtualization it is a key enabler the implementation of automation and orchestration in cloud architectures





- It is the process of presenting a logical view of the physical storage resources to a host computer system, treating all storage media (hard disk, optical disk, tape, etc.) in the enterprise as a single pool of storage.
- It is usually implemented with
  - a network-based approach using Storage Area Networks (SAN) with Fiber Channel or iSCSI protocols for block-based virtualization or
  - via Network Attached Storage (NAS) for file-based virtualization
- Cloud solutions typically offer three types of storage paradigms:
  - Block-based
  - File-based
  - Object-based

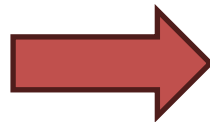
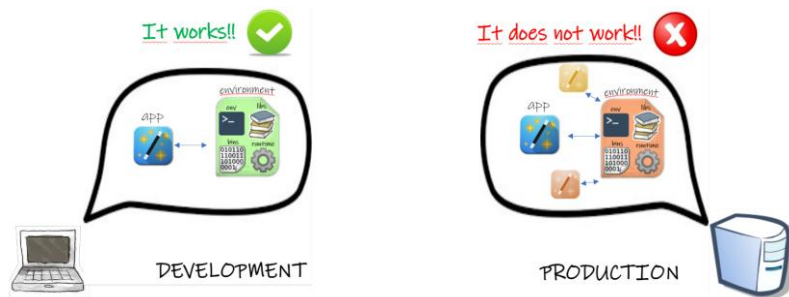
NOTE: A related concept is “Software Defined Storage”, that usually refers to separating the software that carries out core storage operations from physical hardware. Typically, software-defined storage operates on x86-based servers, and turns those servers into storage devices, allowing to aggregate cost-effective storage resources and scale out the system across a server cluster



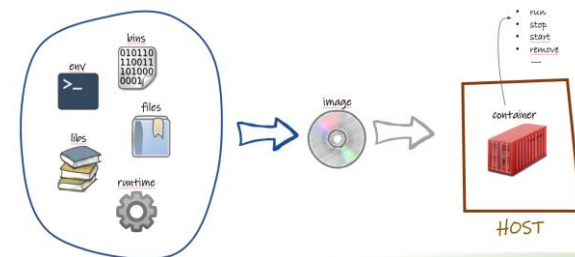


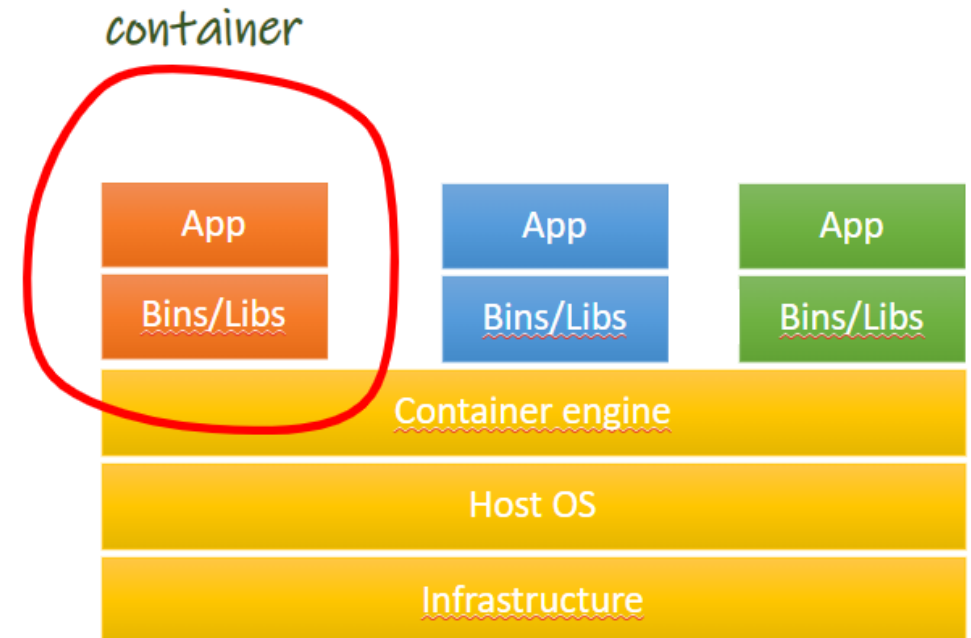
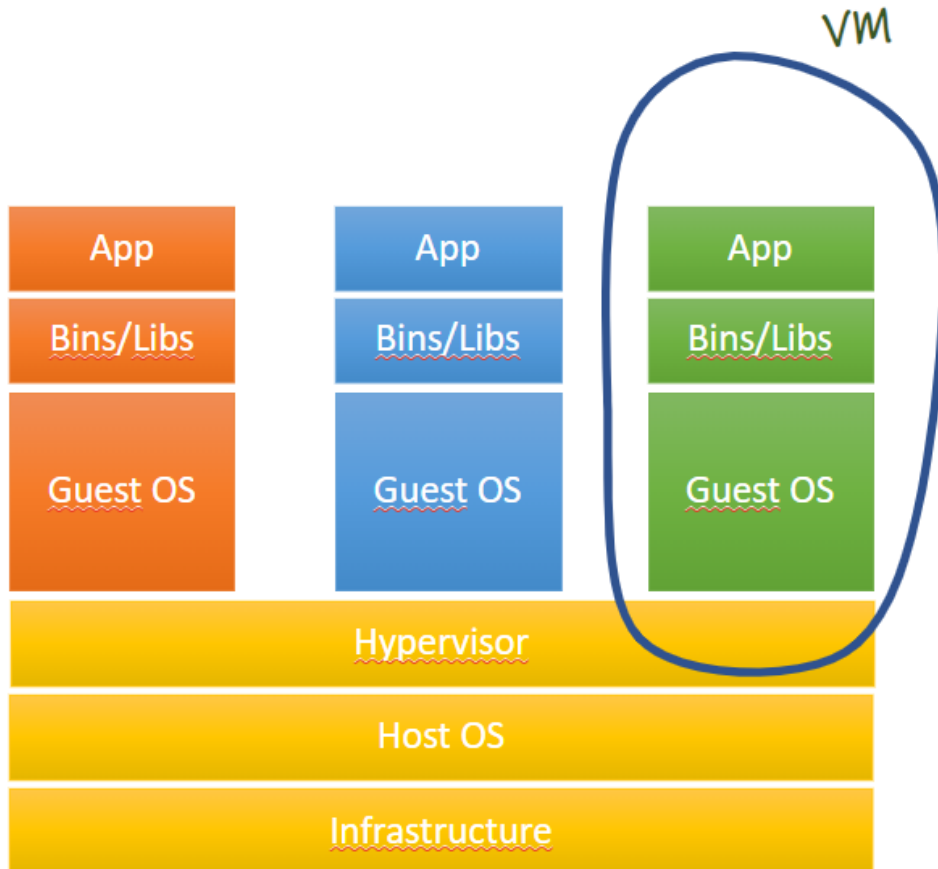
- A container is an isolated runtime instance of an application, defined by its filesystem, configuration, and metadata
- A Docker container image is a lightweight, standalone, immutable, executable package of software that includes everything needed to run an application: code, runtime, system tools, system libraries and settings. Container images become containers at runtime.
- It can be considered a type of operating-system level virtualization.
- Containers require a container runtime to operate within a host system. It is responsible for pulling images from the container registry, managing their lifecycle and running the containers (e.g. docker, podman, crio-o, containerd)
- Containers can provide isolation and encapsulation, and they are much lighter than VMs since they only need to include the SW that applications deployed in them require (no need to include the virtual HW nor the OS kernel)

challenge

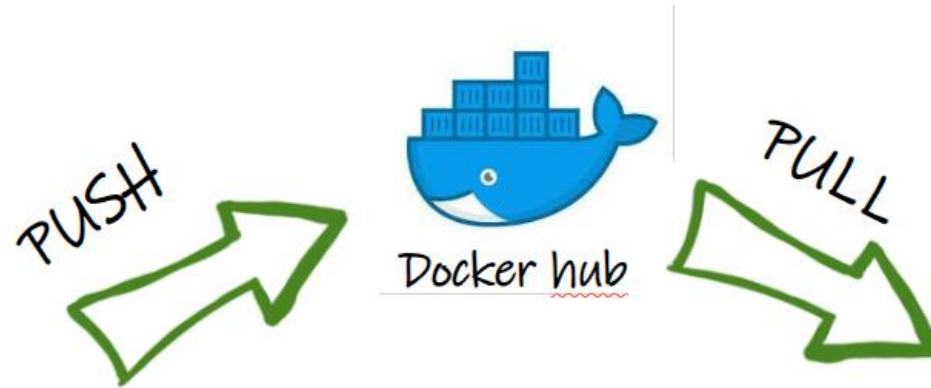


container approach





# Example of build , ship and run with Docker





- A container orchestration platform manages the lifecycles of containers, especially in large, Dynamic environments.
- Kubernetes is an open-source, production.grade, container orchestration platform. It can be used to control and automate a number of tasks:
  - deployment of containers
  - scaling out/in of containers
  - communication between container and with the exterior
  - load balancing and service Discovery
  - health monitoring
  - zero-downtime, easy upgrades
  - allocation of resources to containers
  - storage Management
- Without a container orchestration platform like Kubernetes, large microservices architectures would be very difficult to manage.



# Cloud Platforms

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rmperea@comillas.edu



- The most popular cloud platforms are:
  - Public cloud solutions: Amazon Web Services, Google Cloud, Microsoft Azure, Alibaba Cloud
  - Private cloud platforms: VMWare, Openstack (opensource), Openshift Virtualization
- Features to compare across cloud platforms:
  - Application Programming Interfaces
  - IAM
  - Availability Zones
  - Fault tolerance and failover
  - Elasticity
  - Migration
  - Monitoring, logging, tracing
  - Service Provided



# Orchestration and Automation

Professor: Rogelio Martinez Perea  
rmperea@comillas.edu



- Automation & Orchestration are crucial in a Cloud Computing environment due to its complexity, scale, heterogeneity and constant change involved
- Automation is a must both at Cloud Provider and at Cloud Customer level
- Almost everything is automated in Cloud Computing:
  - Creation and deployment of resources
  - Workload monitoring and Accounting
  - Life Cycle Management, Scaling
  - Monitoring, self-healing
- Declarative, intent-based approaches to automation preferred
- Orchestration refers to the coordination of all the subsystems needed to deploy, upgrade, scale and operate a service
- Infrastructure as Code (IaC) refers to treating the virtualized infrastructure just like the software of an application
  - It provides many benefits in terms of agility to setup and clone environments, maintain consistency of configurations (code can be managed in a version control system) and reduced effort involved
- DevOps integrates development and operations of applications to increase agility, efficiency and quality. This requires automating the whole workflow from development, testing , deployment and operation





# Cloud Security

Professor: Rogelio Martinez Perea  
rmperea@comillas.edu



- Cloud computing introduces or increases certain risks
- Cloud computing have a more complex overall architecture and more complex management systems. This increases the potential attack surface (also applicable to cloud native applications)
- Overall system complexity increases the potential for exploitable configuration errors
- Shared infrastructure across many tenants. The isolation can be disrupted due to SW/HW bugs or misconfigurations
- System management is Internet-facing
- Legal aspects
- Cloud security best practices:
  - Follow general IT security best practices
  - Choose appropriate deployment and service models
  - Access control with MFA
  - Enforce principle of least privilege and separation of duties
  - Assign roles to Groups not to individual users
  - Encrypt data in transit and at rest
  - Encrypt network communication (HTTPS, IPSEC). Network isolation. Utilize IDS/IPS
  - Zero Trust security model
  - Proper Logging and monitoring



Gracias

Professor: Rogelio Martinez Perea  
rmperea@comillas.edu



# BACKUP